CABINET CHEAT SHEET

Power Glove Vision Quick Reference

Current cabinet addresses, services, controls, and maintenance commands.

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This page describes the currently deployed development setup. It intentionally does not contain the private pairing token.

UNO Q

Item	Current value
Board name	`ArduIain`
Network hostname	`arduiain.local`
Current IPv4 address	`10.0.2.105`
Board	Arduino UNO Q
App Lab app	PowerGlove Vision
Camera	Razer Kiyo, configured as `auto`
Configured receiver	`10.0.2.57` (`retropieconsole.local` when mDNS is available)
Startup profile	Super Glove Ball

Prefer the `.local` hostname in bookmarks and configuration. The numeric IP is assigned by the network and may change.

Browser URLs

Purpose	URL
Live debug dashboard	<http://arduiain.local:8088/debug>
Offline gesture lessons	<http://arduiain.local:8088/learn>
Friendly connection setup	<http://arduiain.local:8088/setup>
Secure pairing setup	<https://arduiain.local:8443/setup>
Root (redirects to dashboard)	<http://arduiain.local:8088/>
Machine-readable status	<http://arduiain.local:8088/status>
Camera stream	<http://arduiain.local:8088/stream>
GitHub repository	<https://github.com/mathan416/PowerGlove-Vision>

If `.local` discovery is unavailable, replace `arduiain.local` with the current UNO Q IP address, currently `10.0.2.105`.

Network ports

Port	Protocol	Direction	Use
`8088`	TCP	Browser -> UNO Q	Setup, dashboard, status and camera stream
`8443`	TCP/TLS	Browser -> UNO Q	Secure RetroPie pairing
`55355`	UDP	UNO Q -> RetroPie	Controller-state packets
`55356`	UDP	RetroPie -> UNO Q	Authenticated profile selection and acknowledgement
`55357`	TCP/TLS	UNO Q -> RetroPie	Temporary one-time pairing helper

Keep these ports limited to the trusted local network. Do not expose them to the public internet.

RetroPie defaults

Item	Value
Console hostname	`retropieconsole.local`
Virtual controller	`PowerGlove Vision`
Pairing-token file	`/etc/powerglove/token`
Game registry	`/etc/powerglove/games.json`
Launcher settings	`/etc/powerglove/launcher.json`
Receiver service	`powerglove-receiver.service`
Delayed-start timer	`powerglove-receiver.timer` (45 seconds after boot)
UNO Q shutdown watcher	`powerglove-system-shutdown.path` (`enabled`, `active`)
Shutdown readiness marker	`data/.shutdown-enabled`

The token in `/etc/powerglove/token` must exactly match the token stored in the UNO Q app's `data/device.json`. Never paste that value into GitHub, screenshots, logs or this cheat sheet.

The shutdown watcher, readiness marker, Dashboard and Setup buttons, and API rejection safeguards were verified on September 3, 2026. **Stop controller** leaves the UNO Q running; **Shutdown system** powers Linux off and requires power to be restored or cycled before restart.

Local development files

Purpose	Path
Git working copy	`/Users/mathan/Developer/PowerGlove`
Installation guide	`/Users/mathan/Developer/PowerGlove/docs/INSTALL_README.md`
Gameplay guide	`/Users/mathan/Developer/PowerGlove/docs/GAMEPLAY_GUIDE.md`
Programs A-I reference	`/Users/mathan/Developer/PowerGlove/docs/bad-street-brawler-programs.md`
UNO Q App Lab installation ZIP	`/Users/mathan/Developer/PowerGlove/output/app-lab/PowerGlove-Vision-Uno-Q.zip`
Gesture profiles	`/Users/mathan/Developer/PowerGlove/config/profiles.json`
Automatic game mapping	`/Users/mathan/Developer/PowerGlove/config/games.json`

Deploy over Wi-Fi

The Mac is authorized to maintain the UNO Q using its SSH key. From the local repository, deploy application updates and restart PowerGlove Vision with:

```sh
SH
cd /Users/mathan/Developer/PowerGlove
scripts/deploy-uno-q-wifi.sh
```

One-time installation of the Dashboard and Setup shutdown control:

```sh
SH
cd /Users/mathan/Developer/PowerGlove
scripts/install-uno-q-shutdown-helper.sh arduino@arduiain.local
```

The helper asks for the UNO Q password through the terminal. **Shutdown system** gracefully powers off Linux; restore or cycle power to start the UNO Q again.

This preserves the private `data/device.json`. No password is stored in the repository. USB is only needed as a recovery option or for a passwordless blue matrix firmware upload; App Lab can also perform a wireless firmware update by asking for the board password privately.

Verify key access before deploying:

```sh
SH
ssh -o BatchMode=yes arduino@arduiain.local hostname
```

Pair the current RetroPie

Recommended one-time-code method:

1. On RetroPie, run `sudo /opt/powerglove/bin/powerglove-pair`.

2. Open <https://arduiain.local:8443/setup> on the trusted local network.
3. Enter `retropieconsole.local` and the 20-character code.
4. Prepare pairing and compare the matrix `ID` with the browser certificate's SHA-256 fingerprint prefix.
5. Enter the matrix `PN` digits and complete pairing.
6. Confirm `powerglove-receiver.service` is active.

Username/password pairing is available on the same secure page. Choose **Prepare password pairing** before entering the password, perform the same matrix certificate check, and then complete the one-time SSH operation. The password is not stored.

Quick health checks

Open the status endpoint in a browser or run:

```
SH
curl -sS http://arduiain.local:8088/status
```

Healthy tracking should report:

```
JSON
{
  "camera_available": true,
  "worker_running": true,
  "detected": true,
  "calibrated": true
}
```

Useful RetroPie checks:

```
SH
sudo systemctl status powerglove-receiver.service
sudo systemctl status powerglove-receiver.timer
sudo journalctl -u powerglove-receiver.service -n 100 --no-pager
grep -A8 -B2 'PowerGlove Vision' /proc/bus/input/devices
```

Run the local software tests:

```
SH
cd /Users/mathan/Developer/PowerGlove
PYTHONPATH=src python3 -m unittest discover -s tests -v
```

Camera troubleshooting

The Razer Kiyo must be connected to the UNO Q through an externally powered USB-C hub. Connect to App Lab over Wi-Fi while the UNO Q is acting as the USB host. A camera connected to the Mac is not visible inside the UNO Q app.

If the UNO Q lists only `qcom-venus-encoder` and `qcom-venus-decoder`, the Kiyo is not visible to Linux. Check the powered hub, reconnect the Kiyo, wait several seconds, and watch the dashboard. The supervisor retries automatically; do not cycle through camera indexes when no UVC device exists.

Website images

- `docs/images/debug-dashboard.png`
- `docs/images/learn-page.png`
- `docs/images/setup-page.png`

Profiles

The Dashboard can change the active profile for the current session, and Setup can choose the profile used at startup:

- Bad Street Brawler
- Super Glove Ball
- Power Glove Programs A-I
- Gestures off

RetroPie launch hooks select the registered profile for each recognized ROM. Launching an unregistered game safely selects **Gestures off**. Automatic profile selection currently applies only to NES and Famicom games; launching another system also turns gestures off.

Useful off-script starting points

Program	Good first experiment	Controls and tradeoff
A	Pinball and games built around short bursts	Finger actions suit flippers, wrist movement can suit tilt, and pullback toggles a control. It does not provide an ordinary D-pad.
D	Challenge runs and games where reversed movement is fun	Every direction is reversed; thumb and index actions provide A and B.
H	General NES and Famicom experiments	Hand movement provides a conventional D-pad and thumb/index actions pulse A and B. Pulses are a poor fit for actions that must be held continuously.

Quick off-script test

1. Launch an unregistered NES or Famicom game. Confirm that PowerGlove Vision reports **Gestures off**.
2. Open the UNO Q **Dashboard** and choose Program A, D, H, or another profile.
3. Wait for vision to start and ask for centering.
4. Center your hand, select **Start controller**, and test movement and actions.
5. Stop the controller before changing profiles or games.

This temporary choice is ideal for discovery. It does not change the automatic game registry or the startup profile saved in Setup. While **Gestures off** is selected, the camera and MediaPipe tracker are closed and the UNO Q matrix runs its animated glove attract sequence.

Keep a working combination

Add the ROM's exact basename, including `.nes`, `.zip`, or `.7z`, to the `games` object in `/etc/powerglove/games.json`. Preserve all existing entries. For example:

```json
JSON
{
  "games": {
    "YOUR EXACT GAME FILENAME.nes": "program_h"
  }
}
```

Check the edited file before testing:

```sh
SH
sudo python3 -m json.tool /etc/powerglove/games.json >/dev/null
```

Restart the game so its launch hook selects the newly registered profile. See `docs/GAMEPLAY_GUIDE.md` for the illustrated play and experimentation guide and `docs/bad-street-brawler-programs.md` for complete Program A-I controls.

For Super Glove Ball, hand position controls the D-pad, index curl is A, thumb curl is B, a V sign held for about 0.7 seconds is Start, and a thumbs-up with the other fingers closed is Select.

Startup behaviour

In Arduino App Lab, the current PowerGlove Vision project must show the `DEFAULT` badge. The app then starts automatically after the UNO Q boots. The dashboard becomes available before the camera worker is ready, which makes it the best first place to diagnose startup or camera problems.

On RetroPie, direct boot enablement of `powerglove-receiver.service` is disabled. `powerglove-receiver.timer` starts it after 45 seconds so EmulationStation initializes before the virtual controller appears. This helps prevent frontend pauses and conflicts with other USB devices, including the BitPixel display, when the receiver starts too early.

App Lab currently contains the active `powerglove-vision` installation and an older timestamped import. The older container is stopped. Keep only the active copy set as the default/autostart app; remove the older entry through App Lab after confirming its `data/device.json` is not needed.